Stylized Facts and Multiscaling in Finance – A Literature Review

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St.2 – literature review:

The best documented beginning was probably the work of **Elliot** in the 1930s where he emphasized the **appearance of patterns at different time horizons** (Frost and Prechter 1978).

in order to understand the source of multifractality from an empirical and theoretical point of view: (3)

- Mandelbrot 1963, 1967;
- Mantegna and Stanley 1995, 1999;
- Lux and Marchesi 1999;
- Gençay et al. 2001;
- Calvet and Fisher 2002;
- Lux 2004;
- Di Matteo, Aste, and Dacorogna 2005; Di Matteo 2007;
- Buonocore et al. 2019.

Mandelbrot (**Mandelbrot,1963**) has shown that **financial data** have a **fractal nature**, i.e., that time series of prices in **15 minutes interval have a very similar shape like time series of daily close prices**.

Until the **1960**s, the **only stochastic and scaling model** in finance was the **Brownian motion**, originally proposed by **Bachelier (1900**), and **developed** several decades later (Osborne **1959**).

This theory predicts that the returns of market prices should **follow a normal distribution (**with stable mean and Finite variance). However, There Is Ample empirical evidence that the **returns are not normally distributed**, but have a **higher peak around the mean** and **fatter tails** (Mantegna and Stanley 2000, Dacorogna et al. 2001a, Bouchaud and Potters 2004). (21)

time series gains **a long memory character** when **the return period increases**. Moreover, their distributions move away from the Gaussian normal to so-called **fat-tailed distributions**. **Mandelbrot** (Mandelbrot and Hudson, **2004**) has shown that prices change in financial markets **did not follow a Gaussian distribution**, but rather more general **L´evy stable distributions**.

2001's beef:

- A controversy has erupted between **LeBaron (2001)** on one side and **Mandelbrot (2001) and Stanley and Plerou (2001)** on the other, with **Lux (2001) somewhere in the middle**, to determine if the processes that describe financial Data are truly scaling or if The Similarity observed at different scales is simply an artifact of the data. (21)

Nowadays we now that:

- Multiscaling has been understood to originate from one or more phenomena **related to trading dynamics.**
- In particular, it can be attributed to the **fat tails**, the **autocorrelation of the absolute value of log-returns**, **liquidity dynamics**, **or (non-linear) correlation between high and low returns generated by the different time horizons** at which traders operate and the consequent volumes traded. .

Scaling Properties:

- Similarity S is a **linear transformation**
- two geometrical objects A and B are similar (~) if it exists a couple of similarity transformation S, S' such that S(A) ~= B and S'(B) ~= A.
- S(O) ~= O
- the unitary one dimensional interval is I= [0,1].
- **The fractal object** is therefore defined by:

$$\hat{S}^{\infty}(I)$$

(Hausdorff) dimension: (Fractality dimension)

$$d_F = -\lim_{r \to 0} \frac{\log N(r)}{\log r}$$

- N(r) is the minimum number of d-dimensional balls of radius r

$$d_T < d_F \leq d$$

- Where **d** is its **embedding dimension** (i.e the dimension of the space in which the object lives), **dt** its **topological dimension** (i.e. the local dimension of each point of the object).

In general a differentiable function is said to be scale invariant:

$$f(x) = \lambda^{-\alpha} f(x / \lambda)$$

- **Self-similarity leads to power laws** of the type in above equation but the opposite is not guaranteed.
- In the case of self-affine process:
  one of the most known measure of long-term memory of time series, called the **Hurst exponent**

Typical examples of self-affine processes used in finance are the **L-stable processes** and the **Fractional Brownian Motions (FBM), *BH*(*t*).**

- the FBM has dependent increments with negative autocorrelation for 0<H< 1/2 and positive for 1/2<H<1. When H= 1/2 the FBM becomes a simple Brownian Motion.

Hurst Exponent:

- **autocorrelation function:**

$$C(\Delta t) = E[X(t)X(t+\Delta t)]$$

- if the correlation function behaves as

$$C(\Delta t) \sim 1 - |\Delta t|^{\alpha}, \quad for \quad \Delta t \to 0$$

- for $\alpha \in (0,2]$ one can **extract the fractal dimension** of the random path of $X(t)$ with the relation:

$$d_F = 2 - \frac{\alpha}{2}$$

- analyzing the **asymptotic behaviour at large time lags of $C(\Delta t)$,** one can quantify the **long-range dependence** of the process:

$$C(\Delta t) \sim |\Delta t|^{-\beta}, \quad for \quad \Delta t \to +\beta$$

for $\beta \in (0,1)$, the Hurst exponent is defined from this quantity by the equation

$$H = 1 - \frac{\beta}{2}$$

and gives information about the **long-memory dependence of the process.**

- important to underline that the **fractal dimension is a local property** while the **long-memory dependence is a global one**.

- For **self-similar processes** (in a *n*-dimensional space), the **local properties** are preserved also at larger scales and it follows that:

$$d_F = n + 1 - H$$

In general processes characterized by $H \in (0.5,1)$ are characterized by **long-memory dependence** or **persistence**, while when $H \in (0,0.5)$ the process is said to be **anti-persistent**.

- **Hint:** In general, as it will be shown, **the Hurst exponent can be measured on time series regardless the statistical properties of the underlying process generating them, nevertheless it must be always checked that the assumed scaling relations are empirically valid.**

Stylized facts:

1.Detrending:

When performing statistical analysis of returns, it is important to pay attention to the presence of trends in the data that could **affect the validity of the investigation**. In order to remove this possibility, **the best-fit line (in the least-squares sense**) is removed from the log p(t) time series of each stock, and only after this operation are the returns computed.

$$R_\tau(t) = \log p(t+\tau) - \log p(t)$$

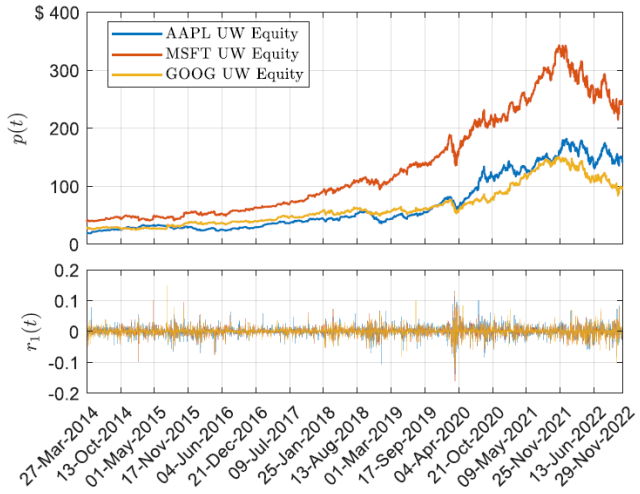


Figure 2.3: Plot of the top 3 most capitalized stocks of the data set: Apple Inc, Microsoft Corp, Alphabet Inc. On the top the price time series, on the bottom the daily return time series. The data is shown from Mar-2014, and not from the beginning of the data set, just to make the plots more clear.

2. Heavy Tails:

Gaussian isn't valid. within the context of risk management, the study of these tails can provide a better understanding of the potential magnitude and likelihood of extreme losses, enabling to make more accurate risk assessments and enhancing the robustness and accuracy of models.

- **excess kurtosis:**

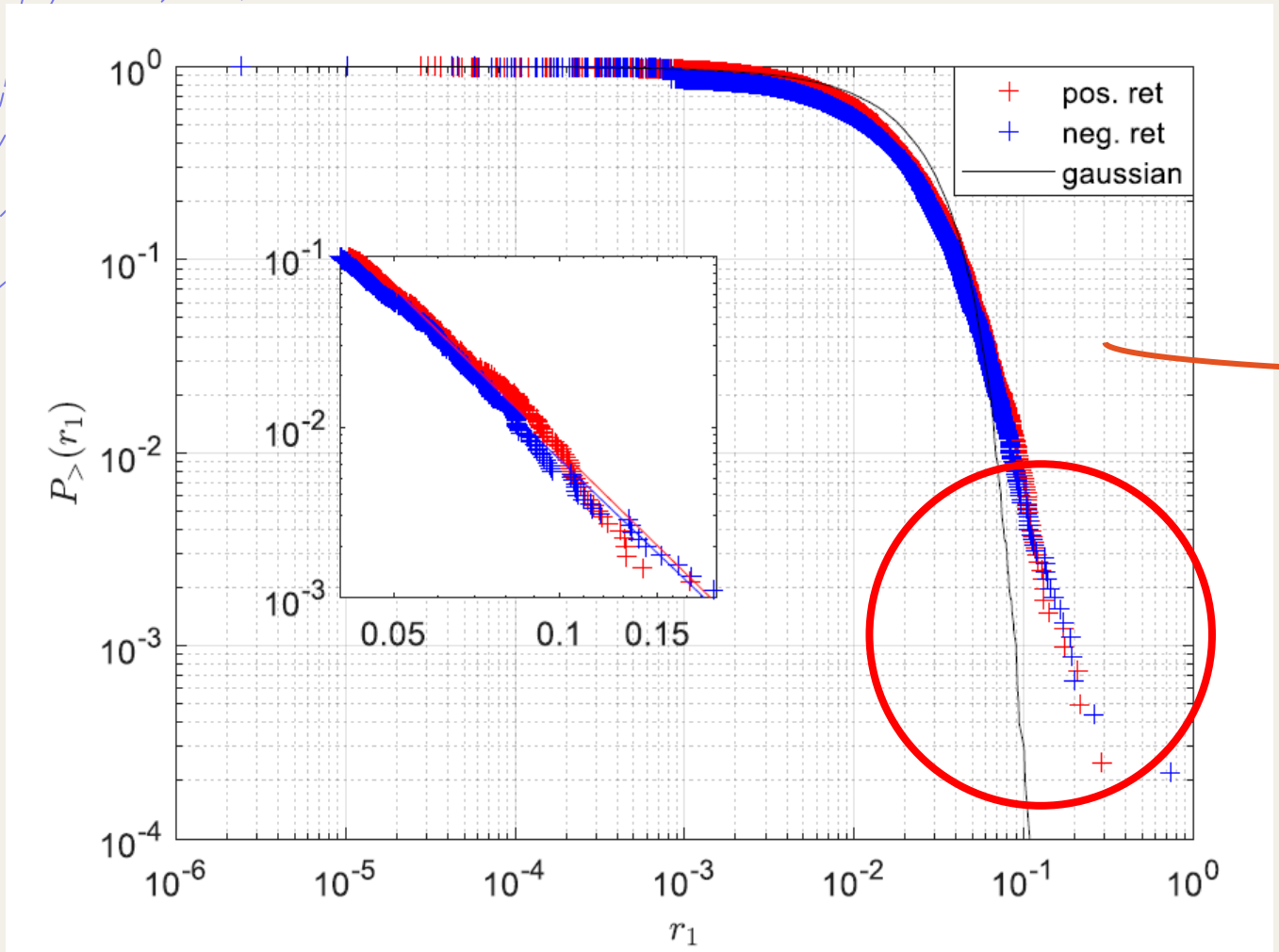
$$\gamma = \frac{\left\langle (r_\tau(t) - \langle r_\tau(t) \rangle)^4 \right\rangle}{\sigma_\tau^4} - 3$$

Knowing that **landau= 0 for a gaussian**, it follows that empirical distributions which show a value higher than 0 are characterized by the so-called **"fat tails"**

almost every time series in the data set is characterized by a **large kurtosis** compared to the gaussian. Distributions with this feature are called ***leptokurtic*** and in financial systems they imply a larger probability to have extreme returns, both positive or negative.

to estimate the tail exponents α of the daily returns of all the stocks in the data set. **Instead of simply applying a least square linear fitting in log-log scale**, it is preferred to **employ a maximum-likelihood fitting method**, which has been proved to provide more reliable results

Using log-log scale to highlight power law behavior:



$$P_>(s;\tau) = \int_s^{+\infty} p(r_\tau)dr_\tau$$

$$P(r_\tau > x) = P_>(x;\tau) \sim x^{-\alpha}$$

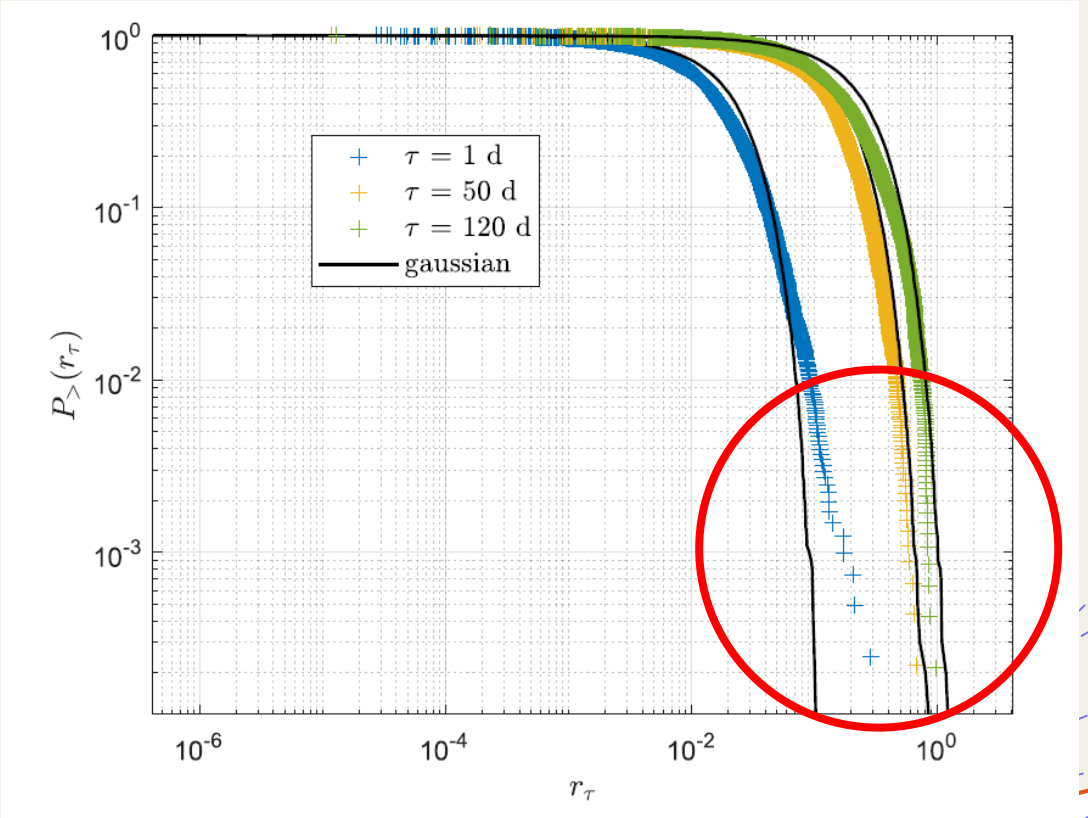
3. Aggregational Gaussianity:

the time scale (tao) over which returns are computed strongly affects the **properties of their distribution.**

When **tao increases**, the heavy tail property of the distribution weakens, and **it approaches a Gaussian form**

- **fat tails, which are less evident when tao increases.**

It is evident that **increasing the time Scale** tao over which returns are computed the distributions get closer to a Gaussian with average and variance computed on the empirical returns.



4. Absence of autocorrelations:

When studying financial data it is fundamental to examine if there exists a **correlation between a given time series and a delayed copy of itself.**

- In this way one can **detect if the system has memory** and therefore if there are **any systematic relationships**, such as **trends** or **patterns**, **between past and future** observations.

Pearson Correlation:

$$C_\tau(\Delta t) = corr(r_\tau(t+\Delta t), r_\tau(t)) = \frac{\mathrm{cov}(r_\tau(t+\Delta t), r_\tau(t))}{\sigma_{r_\tau}(t+\Delta t)\sigma_{r_\tau}(t)}$$

The **autocorrelation immediately decays to zero** and **oscillates around this value for any time lag Δ*t*** employed for the calculations.

- the absence of autocorrelations of returns also acknowledged as an **evidence of the validity of the efficient market hypothesis.**
- **Efficient Market Hypothesis**(Fama, 1970)states that markets are efficient in the sense that the current stock prices reflect completely **all currently known information that could anticipate future market**, i.e., there is no information hidden that could be used to predict future market development. **This model is based on the assumption that market changes can be represented by a normal distribution**.

5. Volatility Clustering:

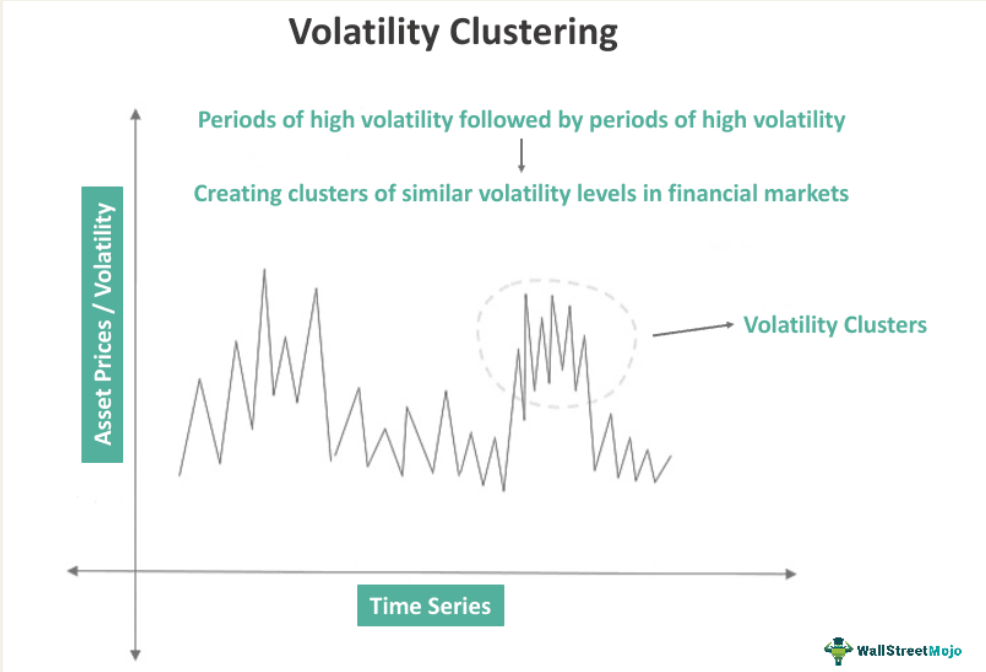
- was first described as the **tendency of financial markets to experience extended periods of high volatility followed by extended periods of low volatility.**

It suggests that **large price movements tend to occur in clusters**, **rather than being randomly distributed over time**, and together with the heavy tails, it is a further evidence that prices cannot be simply modelled as random walks.

- **large returns are often part of a cluster of high volatility while flat regions belong to groups of low price variations.**

- In finance, **volatility clustering** refers to the observation, first noted by Mandelbrot (1963), that:

"large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes."



the autocorrelation of different powers *u* of **the absolute returns** has been proven to approximately decay as a **power law**:

$$\rho(|r_1(t)|^v, |r_1(t+\Delta t)|^v) \sim (\Delta t)^{-\beta}$$

stocks characterized by **low values of the exponent beta** show **longer and more evident clusters of large or small volatility**.

- stocks with **beta ~ 0.1 - 0.2** show a **lot of clusters** throughout the analyzed period and therefore can be considered to be **riskier,** as the price is **more volatile**.
- On the other hand, time series that give back a value of $\boldsymbol{\beta \sim 0.4 - 0.5}$ show **smaller absolute returns and clusters** which **persist for shorter periods**. These last type of stocks are thus characterized by more contained price variations and, in this sense, can be seen as **safer assets**.

different methodologies:

There are **different methodologies** used to compute scaling exponents from time series (Jiang et al. 2019).

Among all, let us recall the **Multifractal Detrended Fluctuation Analysis (MFDFA)** proposed in Kantelhardt et al. (2002), the **Wavelet Transform Modulus Maxima (WTMM)** introduced by Muzy, Bacry, and Arneodo (1991, 1993), the **Deep learning approach** proposed in Corbetta et al. (2021) and the **Structure function approach also known as the Generalized Hurst exponent (GHE)** method (Kolmogorov 1962; Vanatta and Chen 1970; Di Matteo, Aste, and Dacorogna 2003; Di Matteo 2007)

different methodologies:

Multifractal Detrended Fluctuation Analysis (MFDFA):

Which is generalization of the traditional DFA for multifractal analysis of **nonstationary** time series.

Steps of MF-DFA:

1. Determine the **profile**:

- For a given time series x_k of length N, calculate the profile Y(i) as:

$$Y(i) = \sum_{k-1}^{i} (x_k - < x >)$$

Where <x> is the mean of the series, subtraction of the mean is optional as it will later be **detrended**

2. Divide the profile into **segments**:

- Split the profile into **non-overlapping segments** of equal length s.
- If N isn't divisible by s, the remaining part of the series from the opposite end is also divided, resulting in **2N_s** Segments.

HINT: For **short series and negative moments**, the significance of the results for the MF-DFA seems to be slightly better than for the WTMM method

3. **Detrend** each segment:

- For each segment, fit a polynomial to the data and calculate the variance:

$$F^2(s,v) = \frac{1}{s}\sum_{i=1}^{s}[Y((v-1)s+i) - y_v(i)]^2$$

Where y_v(i) is the polynomial fit of order m in segment v.

- Perform this for both sets of segments (starting from the beginning and the end of the profile).

4. Calculate the **fluctuation function**:

- **For each segment**, compute the fluctuation function of **order q**:

$$F_q(s) = \left(\frac{1}{2N_s}\sum_{v=1}^{2N_s}[F^2(s,v)]^{q/2}\right)^{1/q}$$

Where q is a **real parameter**. For **q=2**, this corresponds to the **standard DFA**.

5. Determine the scaling behavior:

- Analyze the log-log plot of F_q(s) vs s. if the time series follows a power-law behavior, then:

$$F_q(s) \sim s^{h(q)}$$

The function **h(q)** is the **generalized Hurst exponent**, which varies with q.

Hint: the generalized Hurst exponent h(q) helps **determine the nature of multifractality**. For **monofractal** time series, h(q) is **independent of q**, whereas **for multifractal** series, h(q) **changes with q**.

different methodologies:

Wavelet Transform Modulus Maxima (WTMM):

analyzing singular signals through the **wavelet transform** and **multifractal formalism**, particularly for **turbulence data**.

1.Wavelet transform (WT):

The wavelet transform is used to **decompose the signal**, enabling analysis in **both physical and scale spaces**. The WT of a signal s(x) is defined as:

$$T_g(a, x_p) = \frac{1}{a}\int s(x) g\left(\frac{x - x_p}{a}\right) dx$$

Here, **a is the scale parameter**, **x_p is the position**, and **g is the analyzing wavelet**.

2. Singularity detection via wavelet modulus maxima:

To detect singularities (**points where the signal behaves irregularly**), the wavelet modulus maxima are **extracted**. These are **local maxima of the absolute value of the WT coefficients**:

$$|T_g(a, x_p)|$$

- **The idea is that singularities of the signal will manifest as maxima in the WT coefficients**.

The authors track these maxima across different scales to **capture the hierarchical structure of singularities**.

3. Partition Function for Multifractal analysis:

Instead of summing all WT coefficients, the partition function is calculated by **summing only the values of the WT modulus maxima**:

$$Z(a,q) = \sum_{x_i(a)} |T_g(a, x_i(a))|^q$$

Here, Z(a,q) captures **the scaling behavior** of the signal across different scales, **for small scales(a --> 0), the partition function follows a power-law behavior.**

$$Z(a,q) \sim a^{\tau(q)}$$

4. Determination of Singularity Spectrum D(h):

The singularity spectrum D(h) provides a **comprehensive description the signals multifractal nature**. It relates the **scaling exponents h** (or **Holder exponents**) to the fractal dimension of the sets of points with singularity strength h.

the relation between D(h) and the scaling exponents tao(q) is found through **Legendre transform**:

$$D(h) = \min_{q}(qh - \tau(q))$$

- Practical Application to **Turbulence**:

This method is applied to turbulence data, where the **multifractal nature of the velocity filed is investigated**. By analyzing the WT modulus maxima of turbulence signals, the authors extract the **Holder exponent h** and the corresponding **singularity spectrum D(h).**

This analysis provides insights into the complex, scale-dependent structures within turbulent flows.

different methodologies:

A Deep learning approach:

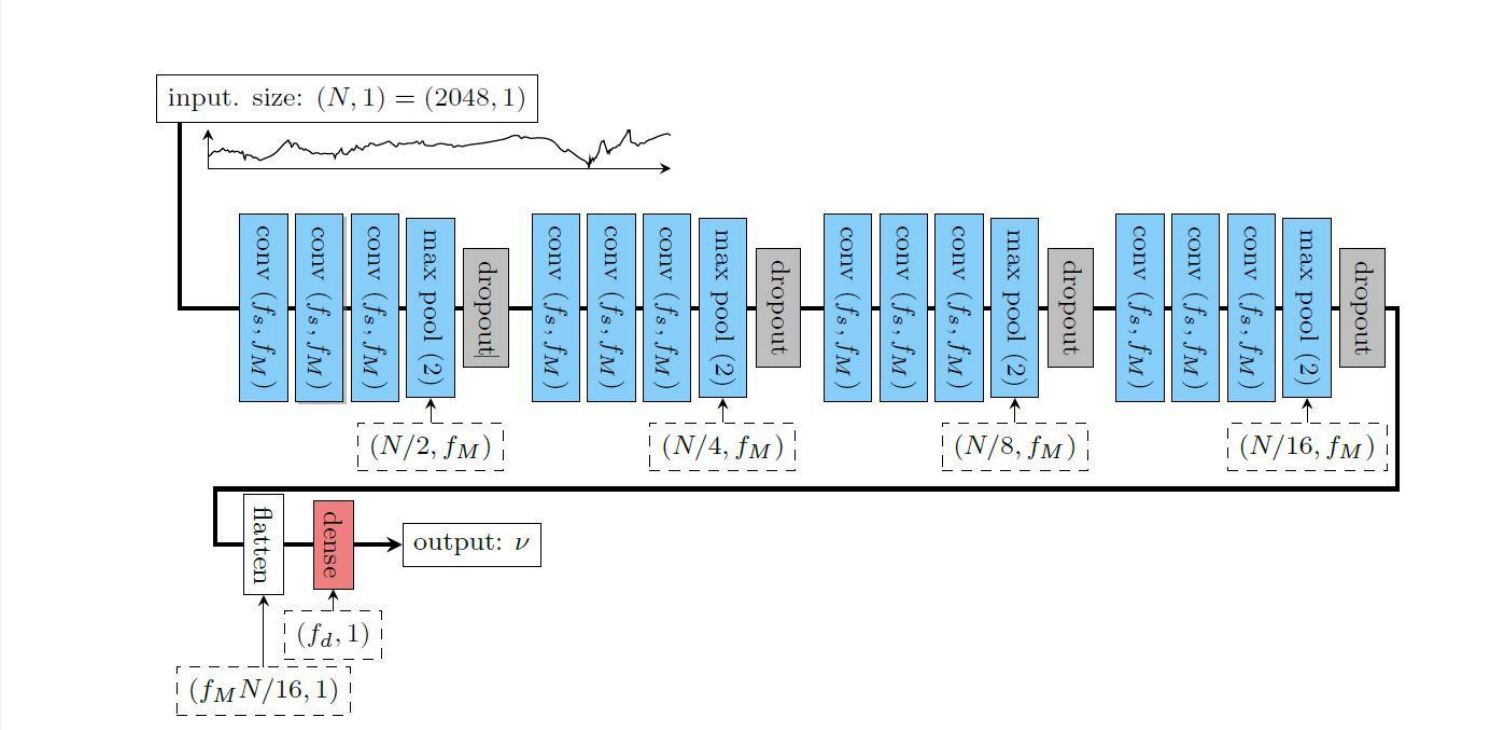
a method to estimate turbulence intensity using deep neural networks (DNNs)

- The paper demonstrates that a deep learning approach can be used to quantify turbulence intensity from **short velocity signals** with remarkable accuracy, even in **nonstationary and complex flows**.

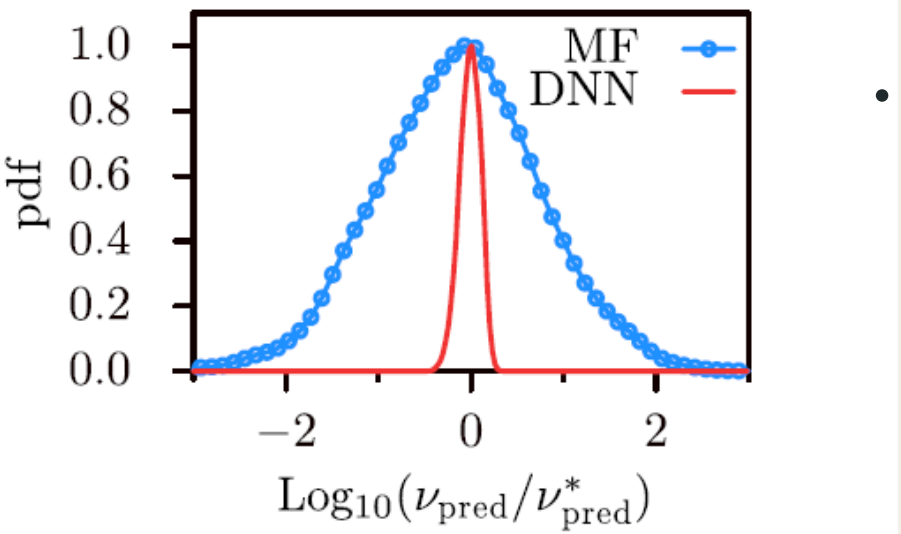
This opens up new possibilities for turbulence analysis, particularly in natural and industrial applications where data is sparse or highly nonstationary.

- The complex space-time patterns and localized correlations in v(t), given by these intermittent bursts, make a **one-dimensional convolutional neural network (CNN)** a well-suited choice for our neural network model

- Comparing **traditional method for turbulence data** and convolutional neural networks (CNN)



- **Feed-forward convolutional neural network** considered. The network is constituted of **four blocks** each encompassing **three convolutional layers** (\conv", filter size fs = 3, filter number fM = 128, activation function: Re-Lu) one max pool layer that down scales the signal by a factor two and, in training, a dropout layer with dropout probability 20%. The dimensions of the feature map as obtained at the end of each block is reported in the dashed rectangles. The last feature map (dimension (128; 128)), is densely connected to a **representation layer** which has fd = 128 dimensions and Re-Lu activation. **The final output, i.e. the predicted viscosity** , is built from a **linear combination of the dense representation values**.



- **Comparison of the viscosity estimates by the DNN [trained through SM data, solid line] and by the MF model (dotted line) for Lagrangian velocity signals obtained by a DNS simulation.**

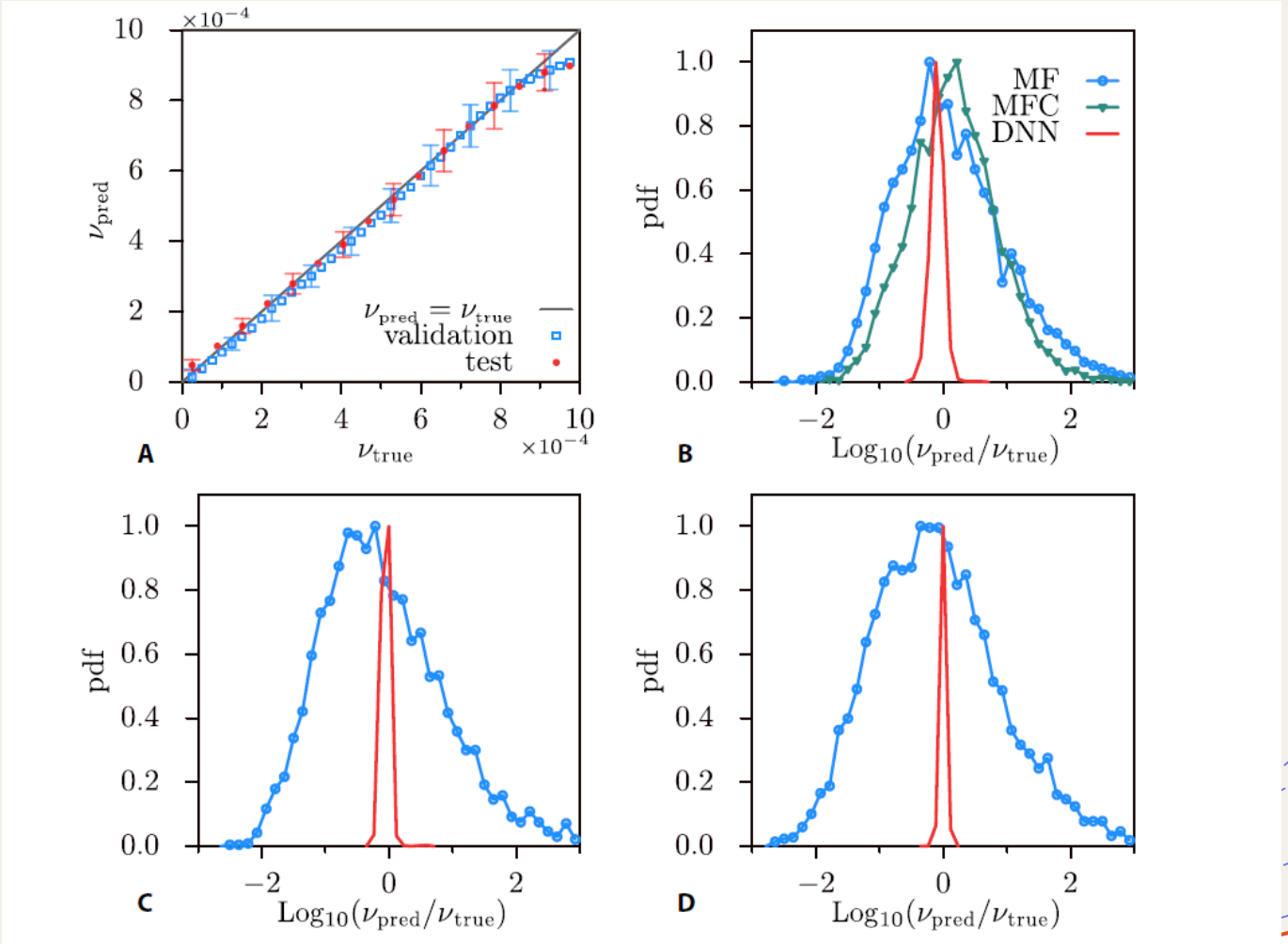
Deep neural network:

We use a one-dimensional CNN architecturally inspired by the **VGG model** (*9*). Developing a neural network model poses the major challenge of selecting a **large number of hyperparameters**. This particular architecture deals with this issue by fixing the size of the filters and uses stacks of convolutional layers to achieve complex detectors. For our model, we opted for **convolutional filters of size 3**, which is **comparable or smaller than the dissipative time scale of the turbulent signals**. The network includes **four blocks**, each formed by **three convolutional layers** (including **128 filters** each), a max pooling layer (**window: 2**), and a **dropout layer**, that capture all the **spatial features of the signal**.

These layers are followed by **a fully connected layer with 128 neurons and Rectified Linear Unit (Re-Lu) activation** that collects all the spatial features into a dense representation. The **final layer** provides a **linear map from the dense representation to the estimated viscosity.**

Lower viscosity values yield higher turbulence intensity,

Neural network viscosity prediction performance and comparison with multifractal-based estimates.



Viscosity estimates via structure functions: traditional method

In the inertial range, **velocity differences**, $\delta v(l) = v(r+l) - v(r)$, exhibit a singular, i.e, **power law behavior** as a function of the observation scale l, i.e.

$$\delta v(l) \sim l^h,$$

For some 0<h<1, let tao_l the typical time-scale at scale l (i.e., $\tau_l \sim l / \delta v(l)$), then it holds

$$\tau_l \sim \frac{1}{\delta v(l)} = l^{1-h},$$

The **dissipative scale** of the system (η, τ_η) satisfies:

$$\frac{\delta v(\eta)\eta}{v} \sim 1,$$

Or, in lagrangian perspective,

From which we have $$\frac{\delta v(\tau_\eta)\tau^{\frac{1}{1-h}}}{v} \sim 1,$$

$$\tau_\eta \sim \nu^{\frac{1-h}{1+h}} \quad and \quad \frac{\delta v(\eta)}{\tau_\eta} \sim \nu^{\frac{2h-1}{1+h}}.$$

Let D(h) be the fractal co-dimension in which an h-singular behavior is observed, i.e. singularities with Holder exponent h occur with probability

$$p(h) \sim l^{3-D(h)} = v^{\frac{3-D(h)}{1+h}}.$$

This enables to express explicitly structure function as

$$\frac{S_2(\tau_\eta)}{\tau_\eta^{\,2}} = \frac{<\delta v(\eta)^2>}{\tau_\eta^{\,2}} = \int dh v^{\frac{3-D(h)}{1+h}} v^{2\frac{2h-1}{1+h}},$$

Which In the fully developed turbulence limit satisfies the power law

$$\frac{<\delta v(\eta)^2>}{\tau_\eta^{\,2}} = v^{-\alpha},$$

Where Alpha is determined by a Legendre transform as:

$$\alpha = \sup_h \frac{D(h)-4h-1}{1+h}.$$

different methodologies:

Structure function approach also known as the Generalized Hurst exponent (GHE) method:

Harold Edwin Hurst

- long-term dependence of **water levels** in rivers and reservoirs
- revealing **long-run correlations** in random processes.

- In a recent paper, Barunik and Kristoufek (2010) tested different methodologies against some data specification and empirically showed that the **GHE approach outperforms the other models.**

Scaling behaviors in differently developed markets – (Di Matteo, 2003)

- **The Robustness of method**
- **Not biased method**

$$K_q(\tau) = \frac{<|X(t+\tau) - X(t)|^q>}{<|X(t)|^q>},$$

Which is a good characterization of the statistical evolution of a stochastic variable X(t).

Note that for q=2, the K_q(tao) is proportional to the autocorrelation function.

The generalized Hurst exponent H(q) can be defined the **scaling behavior of K_q(tao)** which can be assumed to be given by above relation:

$$K_q(\tau) \sim (\frac{\tau}{v})^{qH(q)}.$$

- If H(q) = H → independent of q → uni-scaling or uni-fractal process
- If H=0.5 → SBM = Standard Brownian motion
- If H≠ 0.5 but constant between [0,1] → FBM = fractional Brownian motion
- If H(q) dependent on q → multi-scaling or multi-fractal
- When q=1, H(1) describes the scaling behavior of absolute values of the increments.
- When q=2, H(2) is associated with the scaling of the **autocorrelation function** and is related to the **power spectrum**:

(S(f) ~ f^(-beta). And Beta = 1 + 2H(2)

Methodology:

On **daily time series**

Logarithmic price X(t) = ln(p(t))

- All these variables are '**detrended**' by eliminating a linear drift ηt :

$$< X(t+\tau) - X(t) >= \eta t.$$

- To test the **robustness** of method:

Tao is in the range between v=1 day and tao_max days.

For tao_max: between 5 and 19 days.

- To test that method is **not biased**:

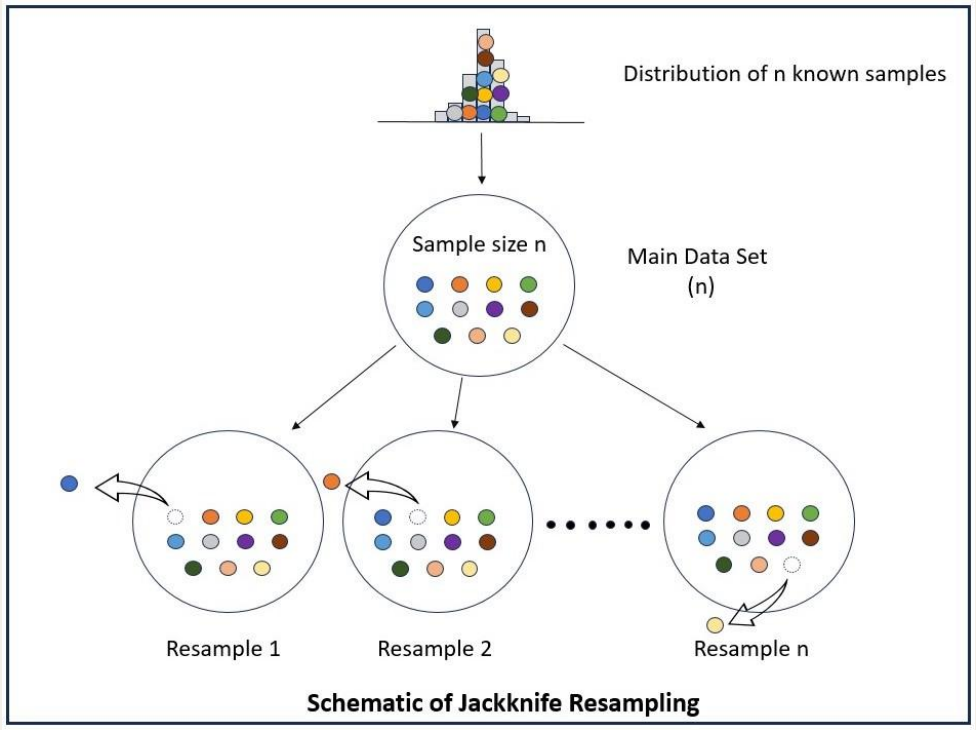
The authors **estimate the GHE** for **simulated random walks** produced by using **3 different random numbers generators**.

- Hint: **in all the cases**, **H(2) have values of 0.5** within the errors.

- **The Method Authors test whether their methodology is “not biased”:**

“Jackknife method”

- **Result: GHE is Sensitive to the degree of market development.**



the scaling law is **not used to conclude anything on the theoretical process** but on the contrary we **use it as a “stylized fact”** that any theoretical model should also reproduce

- What are stylized facts?

A stylized fact is a term **used in economics** to refer to **empirical findings that are so consistent (for example, across a wide range of instruments, markets and time periods) that they are accepted as truth**. Due to their generality, they are **often qualitative**.

Long-term memories of developed and emerging markets: Using the scaling analysis to characterize their stage of development – (Di Matteo – 2005)

- **Developing the last paper idea in more markets** using 2 methods and 5 ways for its **Robustness** and to test which one is "**not biased**"
- Comparing the **GHE** method with **Spectral Analysis** and **R/S Analysis**
- The original Hurst **R/S approach** has **problems** in the **presence of short memory**, **heteroskedasticity**, **multiple scale behaviors**.
- Result: The **Hurst exponent** appears to be **more powerful** in catching the scaling behavior, **even in the frequency domain**.

Scaling exponents in the frequency domain

Spectral analysis: the power spectral density (**PSD**) (Kay and Marple, 1981) by using the **periodogram approach**, that is currently one of the most popular and computationally efficient PSD estimator. This is a sensitive way to estimate the **limits of the scaling regime of the data increments**. As one can see the power spectra show clear **power law** behaviors:

$$S(f) \sim f^{-\beta}$$

- The power spectra coefficient **Beta** are calculated through a mean square regression in log–log scale.

These values differ from the spectral density exponent expected for a **pure Brownian motion** (**Beta = 2**). However, this method is **biased** and we indeed found power **spectra exponents around 1.8 for random walks using three different random numbers generators.**

the autocorrelation function of X(t) is proportional to **the second moment of the distribution** of the increments which, from Eq. (2), is supposed to scale as:

$$K_2 \sim \tau^{2H(2)}$$

But, the **components of the Fourier transform** of a function which behaves in the time domain as

$$\tau^{\alpha}$$

are proportional to

$$f^{-\alpha-1}$$

in the frequency domain. Therefore, we have that the **power spectrum of a signal** that scales as **Eq.(2)** must behave as:

$$S(f) \propto f^{-2H(2)-1}$$

$$K_q(\tau) \sim \left(\frac{\tau}{\nu}\right)^{qH(q)}.$$

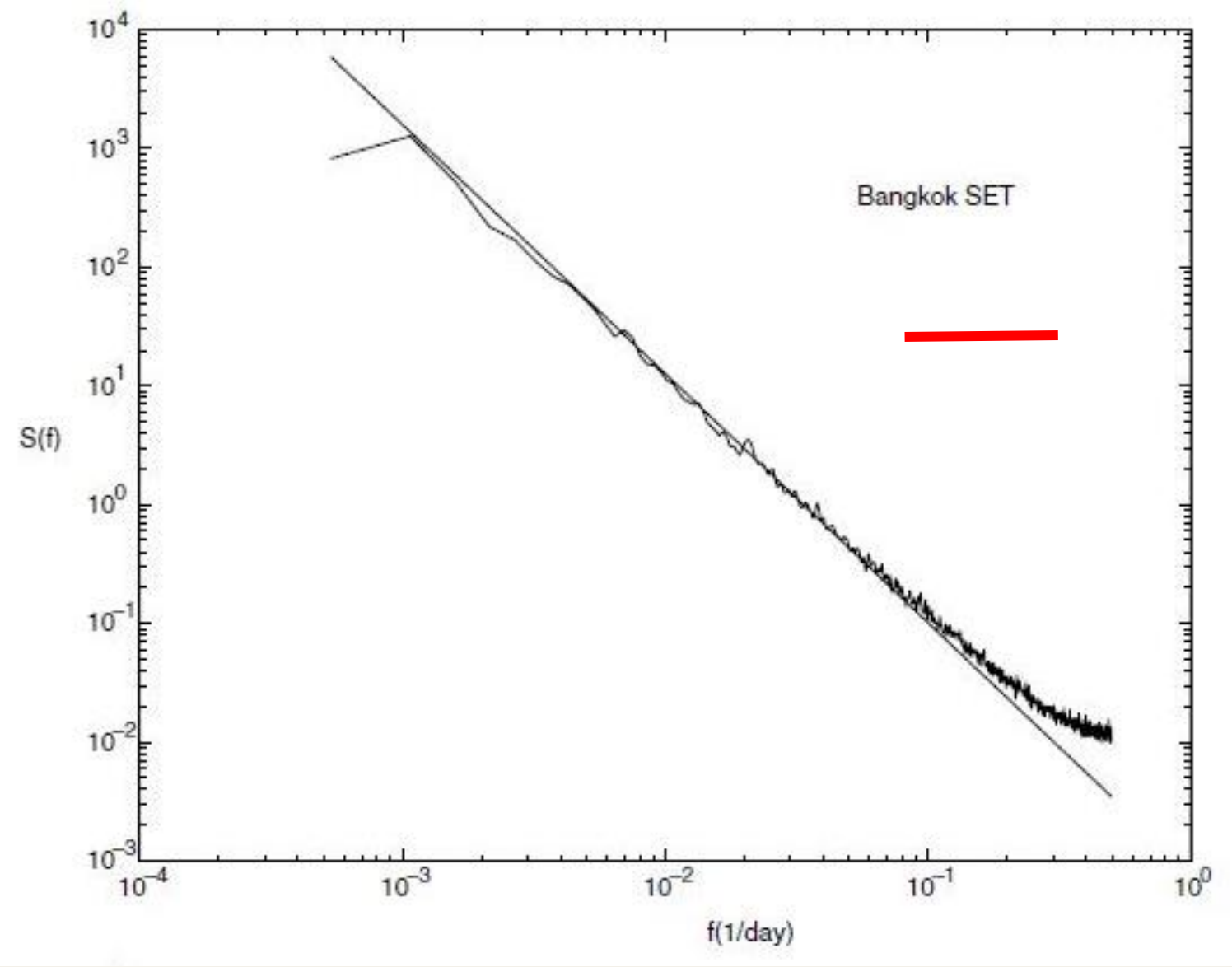
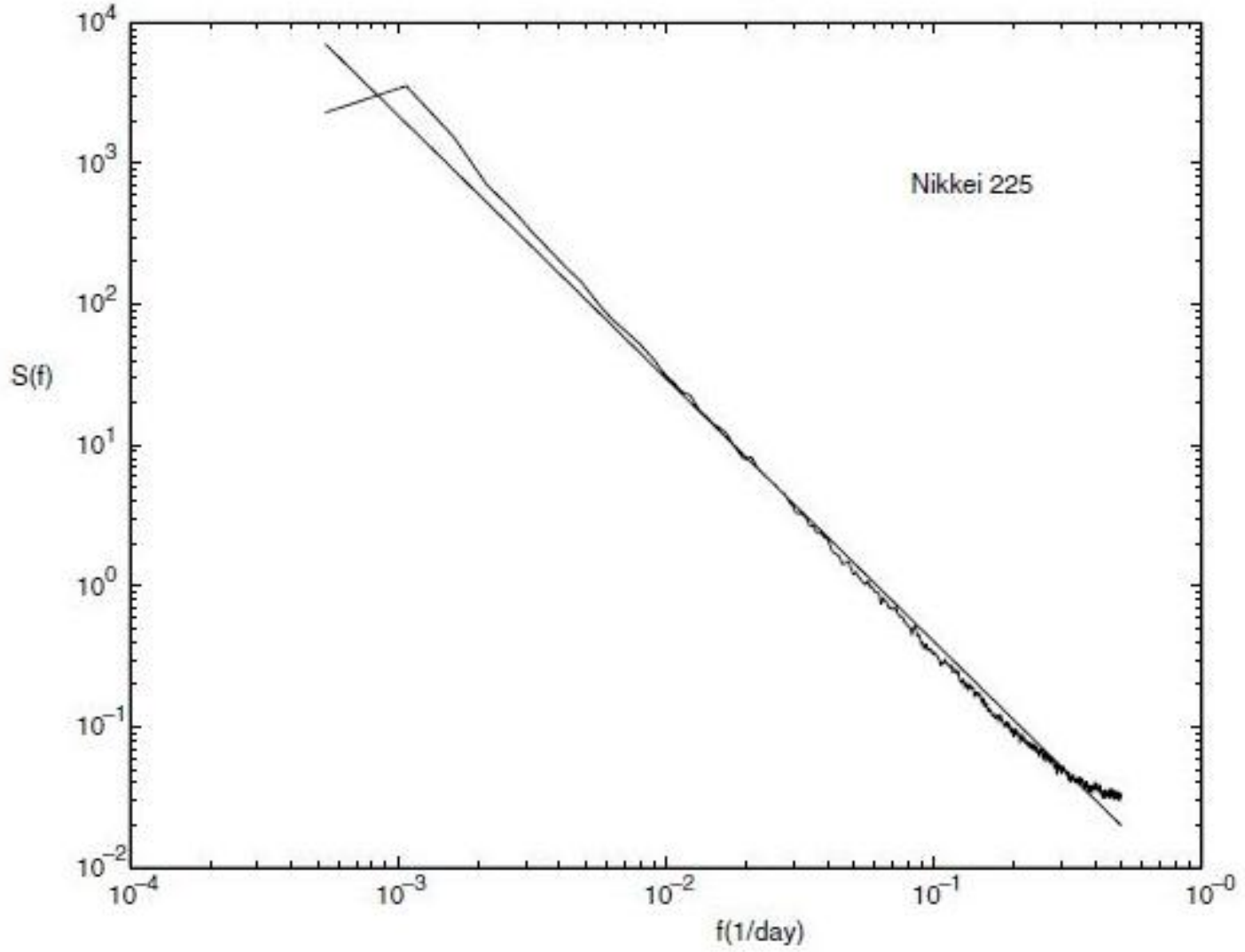


Fig. 6. The **power spectra** of the Stock Market indices compared with the behavior of **f^(-2H(2)-1)** (**straight lines in log-log scale**) computed using **the Hurst exponents** values in the time period 1997–2001:

(a) Thailand (**Bangkok SET**) and (b) JAPAN (Nikkei 225). The line is the prediction from the generalized Hurst exponent H(2) (Eq. (4)).



- It must be noted that, **the power spectrum is only a second-order statistic and its slope is not enough to validate a particular scaling model**: it gives **only partial information about the statistics of the process.**

In order to test that our method is not biased:

we estimate the generalized Hurst exponents for **simulated random walks.(by using monte-carlo)** We produce synthetic time series by using **three different random number generators**. We perform 100 simulations of random walks with the same number of data points as in our samples (991 and 3118) and estimate **the generalized Hurst exponents H(1) and H(2)** and the **power spectra exponents** beta.

- **Results of the test:**

- In all the cases, H(1) and H(2) have values of 0.5 within the errors. Only when we consider **uniformly distributed random numbers** in the interval (0,1) (Rand which uses a **lagged Fibonacci generator** combined with **a shift register random integer generator**, based on the work of Marsaglia (Marsaglia and Zaman, 1994).) we obtain for H(1) of 0.47 ± 0.01, but also in this case H(2) is 0.5 within the errors. On one hand, this shows that **our method is powerful and robust and is not biased as other methods are**.

On the other hand, the **estimations of beta** from the **power spectrum** have values **around 1.8 (instead of 2)**, showing therefore that this other method is affected by **a certain bias**.

The robustness of the present empirical approach is tested in several ways:

by **first** comparing theoretical exponents with the results of **Monte Carlo simulations using three distinct random generators**,

second by varying the **maximum time-step** (Tao_max) in the analysis,

third by applying the **Jackknife method** to produce several samples,

fourth by **varying the time-window sizes** to analyze the **temporal stability** and

fifth by computing results for **detrended and non-detrending** time series.

Multi-scaling in finance – (Di Matteo – 2007):

1.Self-affine process:

A transformation is called affine when it **scales time and size** by **different factors**, while behaviour that reproduces **itself under an affine transformation** is called self-affine (Mandelbrot 1997).

$$X(t) \sim c^{-H} X(ct)$$

for some H > 0 and all c, k, t1, . . . , tk 0 is called self-affine. **H is the self-affinity index, or scaling exponent of the process** {X(t)}.

- **Brownian motion**, the **L-stable process**, and the **fractional Brownian motion** (Feller 1971) are the main examples of self-affine processes in finance.
- empirical evidence shows that many financial series are **not self-affine**, but instead have **thinner tails** and become **less peaked in the bells** when the sampling interval increases(Calvet and Fisher 2002).

2. Multi-scaling process:

A stochastic process {X(t)} is called multi-scaling if it has **stationary increments** and satisfies (Calvet and Fisher 2002)

$$E(|X(t)|^q) = c(q)t^{\tau(q)+1},$$

- all scaling functions must have the same intercept at q=0: Tao(0)=-1. Linear scaling functions (q) are determined by a unique parameter (their **slope**) and the corresponding processes are called **uniscaling or unifractal**.
- Uniscaling processes, which may seem appealing for their **simplicity**, are **not**, however, **satisfactory models for asset returns.** This is because most financial data sets have **thinner tails and become less peaked in the bell** when the sampling interval, delta t, increases.

3. The Holder exponent and the multi-scaling spectrum:

The local Holder exponent, Alpha(t), quantifies the scaling properties of the process at a given point in time, and is also called **the local scale of the process at t**. If we consider a stochastic process X(t), its **infinitesimal variation around time t** is:

$$| X(t+dt) - X(t) | \sim C_t (dt)^{\alpha(t)},$$

where **Alpha(t) and C_t** are, respectively, the **local Holder exponent** and the prefactor at t. The Holder exponent thus describes the **local scaling of a path at a point in time**, and **smaller values correspond to more-abrupt variations.**

- A unique scale **Alpha(t)=1/2** is observed on **the jagged sample paths** of a **Brownian motion**.
- Similarly, a **fractional Brownian process** is characterized by a unique exponent **Alph(t)=H**. The **uniscaling** process is characterized by **one single Holder exponent**, whereas **multi-scaling** processes contain a **continuum of local scales and the Holder exponent is not unique**. In this case, a **multi-fractal spectrum f(Alpha)** can be defined and it can be interpreted as **the limit of the normalized histogram of Holder exponents** (Mandelbrot 1997).

On the statistics of scaling exponents and the multiscaling value at risk – (Di Matteo – 2021)

- **Mathematical approach and the preferment of GHE method**

1. GHE:

This method is a **generalization of the R/S method** to measure the **standard Hurst exponent**, and employs the ***q*-th order moments of the distribution** of the increments of a time series (i.e. returns for financial time series) to extract its **scaling behavior**.

Given $X(t)$ a process with stationary increments, the Generalized Hurst Exponent method (GHE) considers the following function:

$$\Xi(\tau, q) = E[|X(t+\tau) - X(t)|^q] \sim K_q \tau^{qH_q}$$

- that ***qHq*** is a **concave function due to Hölder's inequality.**

- For a standard Brownian motion, which is an uniscaling process, H_q=H=0.5 regardless of q, while processes with **H>0.5** (<0.5) are said to be **persistent** (anti-persistent).

This property can therefore be analyzed performing **a linear regression** in log-log scale:

$$\ln(\Xi(\tau, q)) = qH_q \ln(\tau) + \ln(K_q)$$

the **linearity of the left hand side with respect to ln(*τ*) must be checked**, and if it **holds the method computes different slopes of the straight lines at different *q*** [35]. The so-called **multiscaling proxy** can be devised by fitting the measured scaling exponent with a second degree polynomial [18, 35]:

$$qH_q = Aq + Bq^2$$

It follows that if the empirical **B=0 the process is uniscaling**,

while if **B≠0** the process is **multiscalling**

$$| X(t+\tau) - X(t) | = | \ln P(t+\tau) - \ln P(t) | = | r_\tau(t) |$$

(Tao is always expressed in **days**)

The choice of *q:*

- it is necessary to have $q < \alpha$ where **α is the tail exponent of the return distribution**, otherwise the behavior found by not considering this fact is **severely biased** [8, 11, 18]. Since **it has been empirically shown that financial time series have fat tails with tail exponents ranging from ~ 1.5 to ~ 4** [11, 8], in this project a conservative approach by selecting **$0 < q \leq 1$ is chosen.**

The choice of *τ:*

for daily data $\tau \in$ **[1,19] days**

- approach selecting for daily data $\tau \in [1,19]$ days, which has been proved to be good enough to highlight the multiscaling behavior in financial time series [34]. Nevertheless, this range has **exhibited some biases caused by autocorrelations and power laws** [35]

2. Dynamical Weighted Generalized Hurst Exponent Method:

The average present in Equation ★ in slide 49 is replaced by the following expression:

$$E[f(r_\tau(t))]_\theta = \sum_{s=0}^{\Delta t-1} \omega_s(\theta) f(r_\tau(t-s))$$

- where *f* is **generic function** of the returns $r_\tau(t)$.
- The **weighting factor** w_s can be defined as an **exponentially decaying function of time**:

$$\omega_s(\theta) = \omega_0(\theta) e^{\frac{-s}{\theta}}$$

$$\omega_0 = \omega_0(\theta) = \frac{1-e^{\frac{-1}{\theta}}}{1-e^{\frac{-\Delta t}{\theta}}}$$

$$\sum_{s=0}^{\Delta t-1} \omega_s(\theta) = 1$$

Teta is the **characteristic time**

$$\Xi(\tau, q, \theta) = E[|\, X(t+\tau) - X(t)\,|^q]_\theta \sim K_q \tau^{qH_q^{(\theta)}}$$

- where H_q(teta) is the **wGHE with characteristic time.**
- Regarding the time aggregation **t belongs to [1,19] days** is the **most common** choice.

3. Relative Normalized and Standardized Generalized Hurst Exponent Method (RNGHE and SGHE):

$$\Xi(\tau, q) = E[|\, r_\tau(t)\,|^q] \sim K_q \tau^{qH_q}$$

in order to **reduce the presence of possible biases** introduced in **the estimation**. It is therefore sufficient to **normalize** the function $\Xi(\tau, q)$ as:

$$\tilde{\Xi}(\tau, q) = \frac{\Xi(\tau, q)}{K_q}$$

- to **eliminate the possible bias** introduced by the estimation of *Kq* with **the regression**.

If one then defines the q-order normalized moment as:

$$\dddot{\Xi}(\tau, q_i, q_j) = \frac{\dddot{\Xi}(\tau, q_j)}{\dddot{\Xi}(\tau, q_i)} \sim \frac{\tau^{H_{q_j}}}{\tau}$$

Its scaling relation:

$$\dddot{\Xi}(\tau, q) \sim \tau^{H_q}$$

the multiscaling can be detected by fitting the measured scaling exponent with a **first degree polynomial**:

$$H_q = A + Bq$$

Here **A** is the **linear scaling index** and **B** is the **multiscaling proxy** (different from zero for multiscaling processes)

- To conclude the **relative structure function between two consecutive moments**, namely *q_i* and *q_j (q_j > q_i),* can be defined as follows:

$$\dddot{\Xi}(\tau, q_i, q_j) = \frac{\dddot{\Xi}(\tau, q_j)}{\dddot{\Xi}(\tau, q_i)} \sim \frac{\tau^{H_{q_j}}}{\tau^{H_{q_i}}} = \tau^{H_{q_j} - H_{q_i}} = \tau^{H(q_i, q_j)}, \tag{3.58}$$

where $H(q_i, q_j) = H_{q_j} - H_{q_i}$. Equation (3.58) can also be rewritten in the following way:

$$\begin{bmatrix} \dddot{\Xi}(\tau, 0, q_1) \\ \dddot{\Xi}(\tau, q_1, q_2) \\ \vdots \\ \dddot{\Xi}(\tau, q_{M-1}, q_M) \end{bmatrix} = \tau^{\left[H_{(0,q_1)}, H_{(q_1,q_2)}, \cdots, H_{(q_{M-1},q_M)}\right]}, \tag{3.59}$$

where *M* is the **maximum number of moments used**

- This structure is useful to verify if a process is statistically multiscaling using a **t-test** on each estimated $H(q_i, q_j)$.

- the fact that for uniscaling time series H_q=H implies that the **difference between different order moments is always 0**, except for H(0, q_1).

- it is also possible to use an **F-test** to test if all the coefficients except for the first one ($H(0, q_1)$) are **jointly equal to 0** against the alternative that **some coefficients are different from 0**. This is a **weaker multiscaling test compared to the multiple t-tests**.

- *strongly multiscaling processes* are those processes which **reject both the null hypothesis for all the t-tests and the null of the F-test**;
- *weakly multiscaling processes* are those processes for which **the null hypothesis of all the t-tests is rejected but not the null of the F-test**.
- **These definitions come from the idea that if a process is multiscaling, all the $H(qi, qj)$ are significantly different from 0.**
- However, if the process reconstructed with a **single exponent** is statistically equivalent to the one reconstructed with the full multiscaling spectrum, this means that **such multiscaling behavior is weak**. Finally **if the null hypothesis for one or more t-tests is not rejected but the F-test rejects the null hypothesis**, the process is a *non-stable multiscaling process* [18].

The choice of *τmax:*

In financial time series in fact, there exists a **cutoff** over which the data are uncorrelated and in order to estimate correctly the scaling exponents it is very important **to avoid mixing up the correlated state with the uncorrelated one.**

- **Autocorrelation Segmented Regression (ACSR):**

$$\rho(|r_1(t+\Delta t)|, |r_1(t)|) = \frac{\mathbb{E}[(|r_1(t+\Delta t)| - \mu_1)(|r_1(t)| - \mu_1)]}{\sigma_1^2}$$

Tao_max= (delta)t as the **break point between the correlated state and the random state.**

$$\rho(|r_1(t+\Delta t)|, |r_1(t)|) = \rho(\Delta t) = \begin{cases} \alpha + \beta(\Delta t), & \text{if } \Delta t < \Delta t^* \\ \alpha + \beta(\Delta t^*), & \text{if } \Delta t \geq \Delta t^* \end{cases}$$

Alpha is the **intercept of the regression**

Betta is a **memory exponent for the autocorrelation function**,

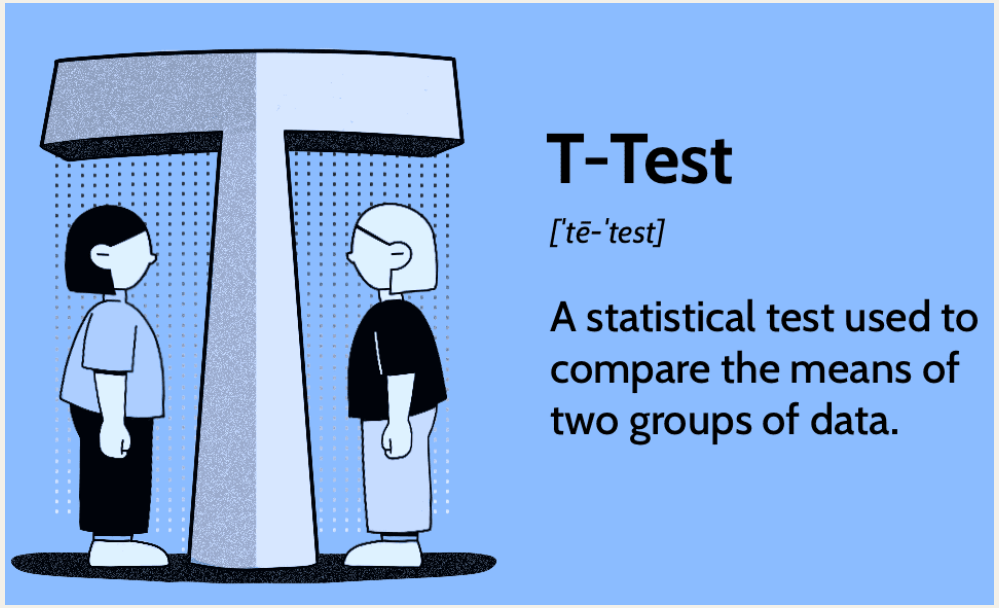
Step by step procedure of RNSGHE:

- Compute τmax
- Fix the vector of q
- Perform the linear regression in log-log scale
- Compute the multiscaling curvature

The test for statistical significance of the results requires the subsequent steps:

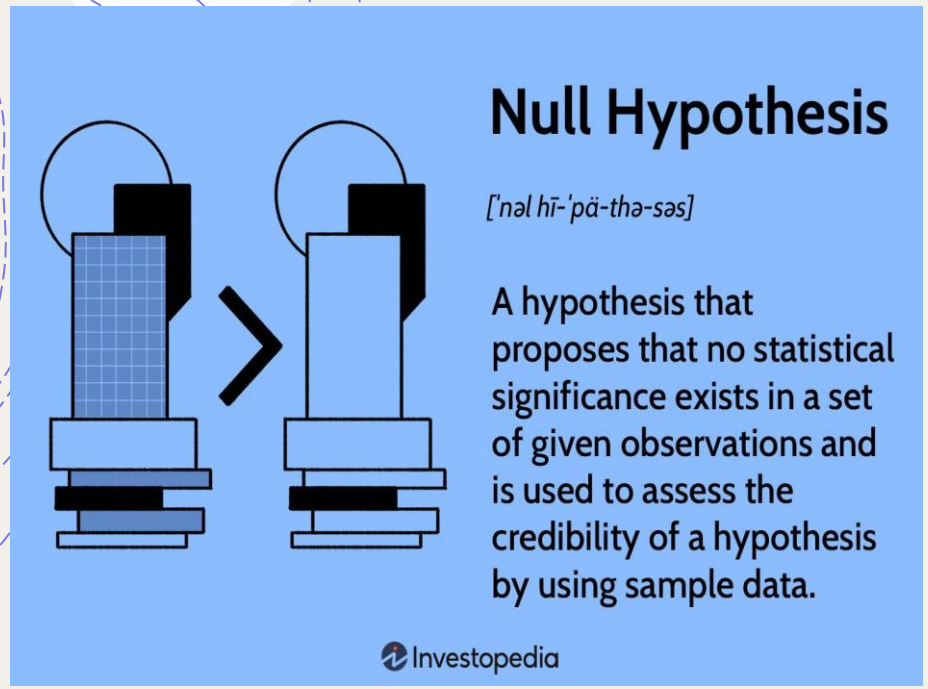
- Test if each scaling increment $H(qi, qj)$ is statistically significant through a t-test, if the null hypothesis is rejected for a coefficient it can be said that it is robustly different from zero.
- Test if all scaling increments except $H(0, q1)$ are jointly different from zero through a F-test, if the null hypothesis is rejected $H(0, q1)$ alone is not enough to describe the scaling behavior.
- Perform the full regression of Equation (3.57) and test for $A = 0.5$ and $B= 01$ using a t-test. If this test gives a conflicting result with respect to the other two steps, a deeper analysis is required.

- A t-test is **an inferential statistic used to determine if there is a significant difference between the means of two groups and how they are related**.
- T-tests are used when the data sets follow a **normal distribution** and have **unknown variances**, like the data set recorded from flipping a coin 100 times.



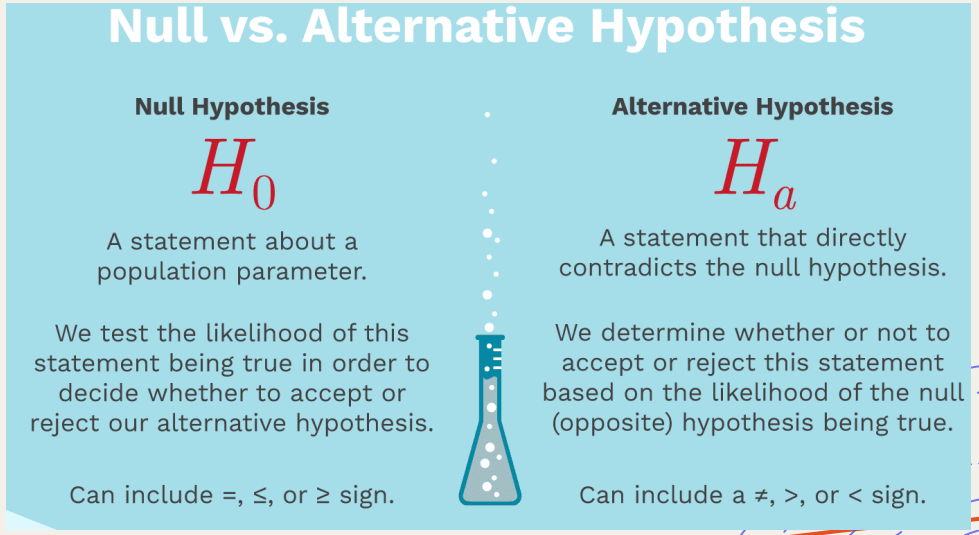
An ***F*-test** is any statistical test used to **compare the variances of two samples or the ratio of variances between multiple samples**. The test statistic, random variable F, is used to determine if the tested data has **an F-distribution under the true null hypothesis**, and true customary assumptions about the error term (ε).

It is most often used when comparing statistical models that have been fitted to a data set, in order to identify the model that best fits the population from which the data were sampled. Exact "*F*-tests" mainly arise when the models have been fitted to the data using least squares.



- **If the null hypothesis is true, it suggests that any changes witnessed in an experiment are because of random chance and not because of changes made to variables in the experiment.**

For example, serotonin levels have no effect on ability to cope with stress.



Some important results:

- **Higher capitalized stocks** tend to be **less multiscaling**.
- **"smaller" assets** show an **higher volatility correlation** which is reflected on the **scaling spectrum.**